

programming languages of today are subsets of LISP. What can be written in one line in LISP may require 3-4 or even more lines in other languages. Course 6.001 uses an academic version of LISP called Scheme. The Structure and Interpretation of Computer Programs is also a text book that is somewhat of a classic in software engineering circles. The entire textbook is available online too, so you have access to the exact same literature as MIT students. And if you are too lazy to read but would rather watch, you can download the video lectures of the course from here: <http://swiss.csail.mit.edu/classes/6.001/abelson-sussman-lectures>.

Online C & C++ courses

www.macs.hw.ac.uk/~rjp/Coursewww/

Any programmer worth his or her salt should understand C and its elder brother, C++. As programming languages go, C and C++ are high level, but basic languages, that are important to gain an understanding of how programming affects the machine. If you are a bit overwhelmed by the MIT course you may skip that and delve right into Professor Pooley's course on C and C++. To effectively use C++ you will need to know C. Both courses are available online and divided into sections. At the end of each section there is a "Programming Milestone"—an exercise segment that will test your grasp of the preceding chapters. After working out the exercises you can check your answers by clicking on the link provided.

Introduction to Programming Using Java

<http://math.hws.edu/javanotes/>

This is a comprehensive course meant for the beginning programmer. It covers the majority of Java 5.0 and will quickly get you up and running. There are exercises, quizzes and solutions with which you can test and grade yourself. You can also download a copy of the entire site for offline browsing. This is especially useful when you don't want to stay connected to the Net for too long.